Experiment no-08

8. Implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model to assign POS tags using python.

PROGRAM:

```python
# Simple Probabilistic POS Tagging

pos_dict = {
    "the": "DT",
    "a": "DT",
    "cat": "NN",
    "dog": "NN",
    "boy": "NN",
    "girl": "NN",
    "runs": "VBZ",
    "eats": "VBZ",
    "is": "VBZ",
    "playing": "VBG",
    "beautiful": "JJ",
    "quickly": "RB",
    "on": "IN",
    "in": "IN"
}

sentence = input("Enter a sentence: ").lower()
words = sentence.split()

print("Word\tPOS Tag")
print("---------------")

for word in words:
    tag = pos_dict.get(word, "UNK")
    print(word, "\t", tag)
```

OUTPUT:

```
Enter a sentence: there is a dog in the way with bone
Word     POS Tag
---------------
there    UNK
is       VBZ
a        DT
dog      NN
in       IN
the      DT
way      UNK
with     UNK
bone     UNK
```